

Soumitra Pal

Curriculum Vitae

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Research Areas

Computational Biology and Bioinformatics, Genomics, Machine Learning, High Performance Computing

Education

Ph.D., Computer Science and Engineering, Indian Institute of Technology – Bombay.

2007-2012 Thesis: Scheduling light-trails on WDM paths and rings: an application of component coloring of graphs.

Advisor: Prof. Abhiram Ranade

M.Tech., Computer Science and Engineering, Indian Institute of Technology – Bombay.

2005-2007 Thesis: Improving Branch-and-Price algorithms for solving one dimensional cutting stock problem.

Advisor: Prof. Abhiram Ranade

B.E., Computer Science & Tech., Indian Institute of Engineering Science & Technology.

1996-2000 Thesis: Synthesis of Cellular Automata and Multiple Attractor Cellular Automata based classification.

Advisor: Prof. Parimal Pal Chaudhuri

Professional Training, Experience and Accomplishments

Staff Scientist, National Eye Institute, National Institutes of Health, USA.

Working with Dr. Anand Swaroop to study the evolution and development of the retina using gene expression data at the bulk and single-cell levels from the retinas of 100+ species spanning the tree of life and organoids.

2022- Key achievements:

- Used Ornstein-Uhlenbeck (OU) process-based methods to identify adaptive expression in nocturnal birds.
- Studied the evolution of the MAF and OTX proteins, key transcription factors for the specification of photoreceptors, before and after two rounds of whole-genome duplication during Cambrian explosion.

Research Fellow, NCBI, National Institutes of Health, USA.

Worked with Dr. Teresa Przytycka to analyze bulk RNA-seq datasets from several organs and species of both sexes of *Drosophila* to identify genes responsible for the evolution of different phenotypic traits. Analyzed sex differences in the Fly Single-Cell Atlas (FCA) in the flagship Science paper as well as several follow up papers.

Key achievements:

- 2019-2022
- Developed EvoGeneX to infer expression evolution: neutral/constrained/adaptive from RNA-seq data.
 - Applied it to multi-species, multi-organ, and multi-sex *Drosophila* data, uncovering adaptive expression.
 - Identified sex-biased cell counts and gene expression in several cell types in FCA.
 - Hypothesized a switch-like role of *dsx* in the regulation of sex-biased gene expression in FCA.

2017-2019 **Postdoctoral Visiting Fellow**, NCBI, National Institutes of Health, USA.

Worked with Dr. Teresa Przytycka to develop a computational pipeline to analyze HT-SELEX data and elucidate the role of DNA shape in Transcription Factor binding.

Key achievements:

- My analysis of the HT-SELEX data showed that there exist distinct motifs of DNA shape in TF binding.
- Developed JUDI, a workflow manager to execute bioinformatics pipelines under varied parameter settings.

Postdoctoral Research Fellow, CSE, University of Connecticut, USA.

Worked with Prof. Sanguthevar Rajasekaran to develop efficient sequential and parallel algorithms for fundamental problems in bioinformatics, including motif search, k-mer counting, and machine learning applications in whole-genome predictions using high-performance computing (HPC).

Key achievements:

- 2014-2017
- Developed an efficient algorithm for finding edit-distance-based motifs in a set of given sequences.
 - Implemented a speedup of parallel Jacobi iterations for computing singular value decomposition (SVD).
 - Developed a hybrid parallel/distributed algorithm for solving machine learning problems using HPC.

Sep-Dec 2015 **Associate Data Scientist**, Nusiv Technologies, USA.

With Prof. Rajasekaran consulted Nusiv on building and using the Knowledge Graph of patient data.

Research Associate, CSE, Indian Institute of Technology – Bombay.

- 2013-2014
- Worked with Prof. Srinivas Aluru and Prof. Abhiram Ranade on a DBT-India funded project on biological sequence analysis using multi-core and many-core architectures.

Key achievement:

- Developed an algorithm for detecting and correcting high-throughput sequencing errors.

Research Intern, IBM India Research Lab, Bangalore.

- May-Aug 2011
- Worked with Dr. T. S. Jayaram to develop algorithms for analytically identifying electric phase each house is connected to the grid using time series data of meter measurements at houses and at transformer sites.

Senior Software Design Engineer, Texas Instruments India, Bangalore.

- 2002-2005
- Designed and implemented several novel features in ISA simulators for TI DSPs: backward-step / backward-run features, save-and-restore, stand-alone loader, and APIs to integrate ISA simulators to different IDEs including TI Code Composer Studio, Virtio Virtual Platform Simulators, Mentor Graphics Seamless CVE.

Software Design Engineer, Texas Instruments India, Bangalore.

- 2000-2002
- Improved performance and extended the ISA simulators for TMS320C54XX series of TI DSPs.

May-July 1999 **Summer Intern**, Digitech India, Kolkata.

Implemented OSPF protocol functions in Digitech's network protocol simulator.

Teaching, Mentoring, and Outreach Activities

- 2026
- Co-mentor**, Graduate Summer Internship Program (SIP), National Institutes of Health (NIH), Bethesda, MD, USA (with Teresa Przytycka).

Mentee: Honray Day, University of Maryland, College Park, MD, USA.

- 2025-26
- Co-mentor**, Alumni Network Program, Indian Institute of Engineering Science and Technology (IIEST), Shibpur, India (conducted on weekends outside NIH duty hours; with Ashish Layek).

- Mentees: Hanzala Sharique, Suraj Kashyap, and Samyajit Das, IEST Shibpur, India.
- 2024-26 **Mentor**, Postbaccalaureate program, National Institutes of Health (NIH), Bethesda, MD, USA.
Mentee: Leo Goubet-McCall, B.S., Pennsylvania State University, PA, USA.
- 2024 **Judge**, 20th Annual Graduate Student Research Symposium, National Institutes of Health (NIH), Bethesda, MD, USA.
- 2023 **Mentor**, High School Summer Internship Program (HS-SIP), National Institutes of Health (NIH), Bethesda, MD, USA.
Mentee: Ojas Chaudhary, Thomas Jefferson School for Science and Technology, VA, USA
- 2023, 26 **Judge**, Fellows Award for Research Excellence (FARE), National Institutes of Health (NIH), Bethesda, MD, USA.
- 2022 **Participant**, five-part seminar series on diversity and inclusion organized by the Office of Intramural Training and Education (OITE), National Institutes of Health (NIH), Bethesda, MD, USA.
- 2020 **Judge**, Postbaccalaureate Poster Presentation, National Institutes of Health (NIH), Bethesda, MD, USA.
- 2020 **Mentor**, Postbaccalaureate Intramural Research Training Award (IRTA) Program, National Institutes of Health (NIH), Bethesda, MD, USA
Mentee: Niko Darby, Sewanee: The University of The South, TN, USA.
- 2018 **Mentor**, Community College Summer Enrichment Program (CCEP), National Institutes of Health (NIH), Bethesda, MD, USA.
Mentee: Mike-Ryan Mofor, Salisbury University, MD, USA.
- 2018 **Participant**, *Scientists Teaching Science* pedagogy course, National Institutes of Health (NIH), Bethesda, MD, USA.
- 2018 Completed a nine-week online course covering learning styles and teaching philosophies, inquiry-based science, curriculum development, classroom dynamics, cultural awareness and diversity, assessment of learning environments, active learning, and development of expert learners.
- Teaching Assistant**, Indian Institute of Technology – Bombay.
 Assisted instructors in evaluating quizzes, exams, assignments, and scribing lecture notes.
- 2005-2009 CS 101: Computer Programming and Utilization Autumn '05
 CS 218/CS 301: Design and Analysis of Algorithms Autumn '07, Spring '09
 CS 435: Linear Optimization Autumn '06
 CS 606: Foundations of Parallel Computing Spring '07, Spring '08

Technical Publications

Book chapters under review/preparation

1. Anderson Mayfield*, **Soumitra Pal**; *Leveraging machine-learning-based text-mining tools for the analysis of 'Omics datasets*; In *Mass Spectrometry in Biomedical Research: From Fundamentals to Advanced Research*; Sanjoy Bhattacharya, Ed.; under review, expected publication August 2027. ISBN: 9780443451263.
2. **Soumitra Pal***, Aleksandr Friman, Anand Swaroop*; *Language and Multimodal Foundation Models in Vision Research: Opportunities from Next-Generation AI*; In *Accelerating vision research with next generation omics and machine intelligence*; Anupam Mondal, Rinki Ratnapriya, Ed.; In preparation, December 2026. (*Co-corresponding)

Manuscripts under review/preparation

3. Leo Goubet-McCall, **Soumitra Pal**, Suja Hiriyanna, Jacob Nellissery, Anand Swaroop*; *Exploring*

- Convergent Mechanisms of Rod Determination in Birds and Mammals*; In preparation, March 2027.
4. Hanzala Sharique, Suraj Kashyap, Samyajit Das, Ashish Kumar Layek*, **Soumitra Pal***; *EvoCluster: Multiscale Leiden Clustering of ESM-C Embeddings for Orthogroup Detection*; In preparation, December 2026. (*Co-corresponding)
5. **Soumitra Pal**, Honray Day, Brian Oliver, Teresa M. Przytycka*; *Sexual dimorphism in Drosophila at single nucleus resolution*; In preparation, December 2026.
6. Mohita Gaur, Anjani Kumari, Madhuri Rudraraju, Nivedita Singh, Matthew Brooks, **Soumitra Pal**, Anand Swaroop*; *Long-term culture reveals transcriptional and 3D genomic remodeling during maintenance of human retinal organoid identity*; In preparation, December 2026.
7. Anjani Kumari, Amy Chen, Kaori Ueda, Alastair Tulloch, Suja Hiriyanna, Jacob Nellissery, **Soumitra Pal**, Anand Swaroop*; *The Rise of the Rod: Evolutionary Reconstruction of NRL Function*; In preparation, December 2026.
8. **Soumitra Pal***, Noor D White Carreiro, Zachary Batz, Anand Swaroop*; *Retinal RNA-seq analysis by phylogenetic Ornstein-Uhlenbeck process-based methods uncover adaptive evolution of gene expression in nocturnal birds*; In preparation, October 2026. (*Co-corresponding)
9. **Soumitra Pal***[†], Noor D White Carreiro[†], Zachary Batz, Leo Goubet-McCall, Caroline Judy, Anand Swaroop*; *Evolution of MAF and OTX families during the emergence and stabilization of rod photoreceptors across Metazoans*; In review, June 2026; doi.org/10.21203/rs.3.rs-10144266/v1. (†Co-first, *Co-corresponding)
10. **Soumitra Pal***, Ojas Chaudhary, Anand Swaroop*; PINNOU: *Physics-Informed Neural Networks for Phylogenetic Ornstein-Uhlenbeck Inference*; In preparation; **Presented two-page extended abstract at EvolCompGen COSI at ISMB**, July 2026; doi.org/10.5281/zenodo.22814087 (*co-corresponding)

Journal Publications

11. **Soumitra Pal**, Jerome Avellaneda, Celena M. Cherian, Puja Biswas, Georg Vogler, Elizabeth J. Rideout, Frank Schnorrer, Teresa M. Przytycka*, Brian Oliver; *Cell type specific allometry controls sex-differences in Drosophila organ size*; Genetics, 233(2), June 2026, iyag091, doi.org/10.1093/genetics/iyag091. **Selected as a GENETICS Featured Article.**
12. **Soumitra Pal***, Brian Oliver, Teresa M. Przytycka*; *Cell-Type Specific Variation in X-Chromosome Dosage Compensation in Drosophila*; microPublication Biology; 2025; doi.org/10.17912/micropub.biology.001501. (*co-corresponding)
13. Laura Campello[†], Matthew Brooks[†], Benjamin Fadl, Hyo Sub Choi, **Soumitra Pal**, Anand Swaroop*; *Transcriptional Heterogeneity and Differential Response of Rod Photoreceptor Pathway Uncovered by Single-Cell RNA Sequencing of the Aging Mouse Retina*; Aging Cell; 2025 Feb 15: e70001; doi.org/10.1111/accel.70001.
14. M.G. Hirsch, **Soumitra Pal**, Farid Rashidi Mehrabadi, Salem Malikic, Charli Gruen, Antonella Sassano, Eva Pérez-Guijarro, Glenn Merlino, Cenk Sahinalp, Erin K. Molloy, Chi-Ping Day*, Teresa M. Przytycka*; *Stochastic modelling of single-cell gene expression adaptation reveals non-genomic contribution to evolution of tumor subclones*; Cell Systems, 2025; 16(1); doi.org/10.1016/j.cels.2024.11.013.
15. Sharvani Mahadevaraju*[†], **Soumitra Pal***[†], Pradeep Bhaskar, Brennan McDonald, Leif Benner, Luca Denti, Davide Cozzi, Paola Bonizzoni, Teresa M. Przytycka, Brian Oliver*; *Diverse somatic Transformer and sex chromosome karyotype pathways regulate gene expression in Drosophila gonad development*; eLife 2024; doi.org/10.7554/eLife.101641. (†Co-first; *co-corresponding)
16. Raghuveer Kavarthapu*, Rajakumar Anbazhagan, **Soumitra Pal**[#], Maria L. Dufau*; *Single-Cell Transcriptomic Profiling of the Mouse Testicular Germ Cells Reveals Important Role of Phosphorylated GRTH/DDX25 in Round Spermatid Differentiation and Acrosome Biogenesis*

- during Spermiogenesis; International Journal of Molecular Sciences 24, no. 4: 3127 (2023); doi.org/10.3390/ijms24043127. (#main bioinformatician; journal did not permit co-first among 4 authors)
17. Amelie Raz[†], Gabriela Vida[†], Sarah Stern[†], Sharvani Mahadevaraju[†], Jaclyn Fingerhut[†], Jennifer Viveiros[†], **Soumitra Pal**[†], Jasmine Grey[†], Mara Grace[†], Cameron Berry, Hongjie Li, Jasper Janssens, Wouter Saelens, Zhantao Shao, Chun Hu, Yukiko M Yamashita, Teresa Przytycka, Brian Oliver, Julie A Brill, Henry Krause, Erika Matunis, Helen White-Cooper, Stephen DiNardo*, Margaret T Fuller*; *Emergent dynamics of adult stem cell lineages from single nucleus and single cell RNA-Seq of Drosophila testes*; eLife 12:e82201 (2023); doi.org/10.7554/eLife.82201. (†Co-first)
 18. **Soumitra Pal**, Brian Oliver, Teresa M. Przytycka; *Stochastic Modeling of Gene Expression Evolution Uncovers Tissue- and Sex-Specific Properties of Expression Evolution in the Drosophila Genus*; Journal of Computational Biology; 2022; doi.org/10.1089/cmb.2022.0121. **ASBMB Best Poster Award to Soumitra; Rosalind Franklin Society Special Award in Science to Teresa.**
 19. Hongjie Li, Jasper Janssens, ..., **Soumitra Pal**, Teresa M. Przytycka, ..., Brian Oliver, Norbert Perrimon, Bart Deplancke, Stephen R. Quake, Liquan Luo, Stein Aerts; *Fly Cell Atlas: a single-cell transcriptomic atlas of the adult fruit fly*; Science 375, no. 6584 (2022); eabk2432; doi.org/10.1126/science.abk2432
 20. **Soumitra Pal**, Tingyang Xu, Tianbao Yang, Sanguthevar Rajasekaran, Jinbo Bi; *Hybrid-DCA: A Double Asynchronous Approach for Stochastic Dual Coordinate Ascent*; Journal of Parallel and Distributed Computing; Volume 143, September 2020, Pages 47–66; doi.org/10.1016/j.jpdc.2020.04.002
 21. **Soumitra Pal**, Teresa M Przytycka; *Bioinformatics pipeline using JUDI: Just Do It!*; Bioinformatics; Volume 36, Issue 8, 15 April 2020, Pages 2572–2574; doi.org/10.1093/bioinformatics/btz956
 22. Subrata Saha, Jethro Johnson, **Soumitra Pal**, George M Weinstock, Sanguthevar Rajasekaran; *MSC: a metagenomic sequence classification algorithm*; Bioinformatics; Volume 35, Issue 17, 1 September 2019, Pages 2932–2940; doi.org/10.1093/bioinformatics/bty1071
 23. **Soumitra Pal**, Jan Hoinka, Teresa M Przytycka; *Co-SELECT reveals sequence non-specific contribution of DNA shape to transcription factor binding in vitro*; Nucleic Acids Research; Volume 47, Issue 13, 26 July 2019, Pages 6632–6641; doi.org/10.1093/nar/gkz540
 24. Peng Xiao, **Soumitra Pal**, Sanguthevar Rajasekaran; *Randomised sequential and parallel algorithms for efficient quorum planted motif search*; International Journal of Data Mining and Bioinformatics; Volume 18, No. 2, pages 105–124; doi.org/10.1504/IJDMB.2017.086457.
 25. **Soumitra Pal**, Peng Xiao, Sanguthevar Rajasekaran; *Efficient sequential and parallel algorithms for finding edit distance based motifs*; BMC Genomics; Volume 17, Article No. 465, 18 August 2016; doi.org/10.1186/s12864-016-2789-9
 26. Abdullah-Al Mamun, **Soumitra Pal**, Sanguthevar Rajasekaran; *KCMBT: a k-mer Counter based on Multiple Burst Trees*; Bioinformatics; Volume 32, Issue 18, 15 September 2016, Pages 2783–2790; doi.org/10.1093/bioinformatics/btw345.
 27. **Soumitra Pal**, Srinivas Aluru; *In search of perfect reads*; BMC Bioinformatics; Volume 16, Article No. S7, 7 December 2015; doi.org/10.1186/1471-2105-16-S17-S7
 28. Ajit Diwan, **Soumitra Pal**, Abhiram Ranade; *Fragmented coloring of proper interval and split graphs*; Discrete Applied Mathematics; Volume 193, 1 October 2015, Pages 110–118; doi.org/10.1016/j.dam.2015.04.014
 29. **Soumitra Pal**, Abhiram Ranade; *Scheduling light-trails on WDM rings*; Journal of Parallel and Distributed Computing; Volume 72, Issue 10, October 2012, Pages 1226–1236; doi.org/10.1016/j.jpdc.2012.05.010

Refereed Conference Publications

30. **Soumitra Pal**, Sudipta Pathak, Sanguthevar Rajasekaran; *On Speeding-up Parallel Jacobi Iterations for SVDs*; IEEE International Conferences on High Performance Computing and Communications; Sydney, NSW, 2016, pages 9–16; doi.org/10.1109/HPCC-SmartCity-DSS.2016.0013
31. Peng Xiao, **Soumitra Pal**, Sanguthevar Rajasekaran; *qPMS10: A Randomized Algorithm for Efficiently Solving Quorum Planted Motif Search Problem*; IEEE International Conference on Bioinformatics and Biomedicine; Shenzhen, 2016, pages 670–675; doi.org/10.1109/BIBM.2016.7822598
32. **Soumitra Pal**, Sanguthevar Rajasekaran; *Improved Algorithms for Finding Edit Distance Based Motifs*; IEEE International Conference on Bioinformatics and Biomedicine; Washington DC, 2015, pages 537–542; doi.org/10.1109/BIBM.2015.7359740
33. Abdullah-Al Mamun, **Soumitra Pal**, Sanguthevar Rajasekaran; *Efficient techniques for k-mer counting*; IEEE International Conference on Computational Advances in Bio and Medical Sciences; Miami, FL, 2015, pages 1-1; doi.org/10.1109/ICCABS.2015.7344733
34. **Soumitra Pal**, Srinivas Aluru; *In Search of Perfect Reads*; IEEE International Conference on Computational Advances in Bio and Medical Sciences; Miami, FL, 2014, pages 1–2; doi.org/10.1109/ICCABS.2014.6863919
35. Vijay Arya, T. S. Jayram, **Soumitra Pal**, Shivkumar Kalyanaraman; *Inferring Connectivity Model from Meter Measurements in Distribution Networks*; International Conference on Future Energy Systems; Association for Computing Machinery, New York, NY, USA, 173–182; doi.org/10.1145/2487166.2487186
36. **Soumitra Pal**, Abhiram Ranade; *Scheduling Light-trails on WDM Rings*; International Conference on Advanced Computing and Communications; December 2009; arxiv.org/abs/1112.6254

Other Manuscripts

37. T. S. Jayram, **Soumitra Pal**, Vijay Arya; *Recovery of a Sparse Integer Solution to an Underdetermined System of Linear Equations*; NIPS Sparse Representation and Low-rank Approximation Workshop; Granada, Spain, December 2011; arxiv.org/abs/1112.1757
38. **Soumitra Pal**, Ish Dham, Tor E. Jeremiassen; *Design and Validation Techniques to Implement a Robust Hindsight Feature on ISA Simulators*; Global Signal Processing Expo and Conference; Santa Clara, USA. September 2004
39. **Soumitra Pal**, Prem Kumar Vadapalli; *A methodology for Saving and Restoring the State of ISA simulators*; Global Signal Processing Expo and Conference; Santa Clara, USA. September 2004

Awards and Honors

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| 2024 | Richard Newton Lolley Memorial Travel Grant, Association for Research in Vision and Ophthalmology (ARVO), Seattle, USA |
| 2022 | Best poster award at ASBMB symposium on Evolution and Core Processes in Gene Expression, Kansas City, USA |
| 2018 | NIH Summer Research Mentorship Award. |
| 2014-16 | Postdoctoral Research Fellowship, University of Connecticut. |
| 2011 | Microsoft Research India Travel Award for presenting at NeurIPS conference, Spain. |
| 2011 | Best Project Award (out of about 100 interns), IBM India Research Lab. |
| 2007-12 | Doctoral Research Scholarship, Ministry of Human Resource Development, India. |
| 2005-07 | Master of Technology Scholarship, Ministry of Human Resource Development, India. |
| 2005 | Secured 49th position in GATE (entrance test for graduate courses at IITs) |

2000 Runners up, IDEON Idea Presentation Competition, IIT – Kharagpur, India

Grants

- 2025 Cloud-Based Training of Large Language Models for Retinal Transcriptome Analysis. NIH Office of Data Science Strategy. **Support for exploratory use of the cloud** via the Science and Technology Research Infrastructure for Discovery, Experimentation, and Sustainability (STRIDES) Initiative FY2025. Program Official: Anand Swaroop. Project Lead: Soumitra Pal. \$200,000.
- 2016 Hybrid Computational Techniques for Genome-Wide Association Study. PI: Soumitra Pal, NHGRI/NIH K99/R00, submitted in October 2016, was reviewed and received a score.

Talks and Presentations

- 2026 [Poster] Evolution of MAF and OTX families during the emergence and stabilization of rod photoreceptors across metazoans, NIH Research Festival, Bethesda, USA.
- 2026 Physics-Informed Neural Networks for Phylogenetic Ornstein–Uhlenbeck Inference, Thirty-fourth Conference on Intelligent Systems for Molecular Biology (ISMB), Washington DC, USA.
- 2025 [Poster] Phylogenomic analysis reveals evolutionary origins and divergence of maf proteins through whole genome duplications. Annual Meeting of the Association for Research in Vision and Ophthalmology (ARVO), Salt Lake City, Utah, USA.
- 2024 Adaptive evolution of gene expression in nocturnal birds. NNRL Symposium 2024, NEI, NIH, Bethesda, MD, USA.
- 2024 [Poster] Retinal RNA-seq analysis by phylogenetic Ornstein-Uhlenbeck process-based methods uncover adaptive evolution of gene expression in nocturnal birds. Annual Meeting of the Association for Research in Vision and Ophthalmology, ARVO 2024. **Richard Newton Lolley Memorial Travel Grant Award.**
- 2023 Evolution of retinal cell-types and retinal adaptation for nocturnality. NNRL Symposium 2023, NEI, NIH, Bethesda, MD, USA.
- 2022 [Poster] Stochastic modelling of gene expression evolution uncovers tissue and sex specific properties of expression evolution in the Drosophila genus. ASBMB Symposium on Evolution and Core Processes in Gene Expression. **Best Poster Award.**
- 2022 [Poster] Dynamic time warping on sn- and sc-RNA-seq trajectories of Drosophila adult and larvae testis enables contrasting the different germline developmental stages. Annual Drosophila Research Conference (Dros22).
- 2021 [Poster] Modeling gene expression evolution with EvoGeneX uncovers differences in evolution of species, organs and sexes. The Biology of Genomes meeting (virtual). Cold Spring Harbor Laboratory, New York, USA.
- 2021 [Virtual talk] Gene expression evolution across species, organs, and sexes in Drosophila. Workshop on Computational Biology, International Conference on Machine Learning conference (ICML).
- 2021 [Virtual poster] Modeling gene expression evolution with EvoGeneX uncovers differences in evolution of species, organs, and sexes. Annual Drosophila Research Conference (Dros21).
- 2020 [Virtual talk] Stochastic modeling of gene expression evolution uncovers differences in evolution of species, organs, and sexes in Drosophila. Intelligent Systems for Molecular Biology Conference (ISMB), Montreal, Canada.
- 2020 Stochastic modeling of gene expression evolution uncovers differences in evolution of species, organs, and sexes in Drosophila. Laboratory of Biological Modeling, NIDDK, NIH, Bethesda, MD, USA.
- 2018 [Poster] Novel Analysis of HT-SELEX Data Elucidates the Role of DNA Shape in Transcription Factor Binding. Intelligent Systems for Molecular Biology Conference (ISMB), Chicago, IL, USA.

- 2017 [Poster] Analysis of HT-SELEX Data Shows Role of DNA Shape Readout in Sequence Non-specific Binding of Transcription Factors. National Institutes of Health Research Day, Bethesda, MD, USA.
- 2016 Some Novel Sequential and Parallel Algorithms for Biological Data Analytics. National Center for Biotechnology Information, NIH, Bethesda, MD, USA.
- 2016 Some Novel Sequential and Parallel Algorithms for Biological Data Analytics. The Jackson Laboratory, Farmington, CT, USA.
- 2015 Efficient Algorithms for Finding Edit Distance Based Motifs. IEEE International Conference on Bioinformatics and Biomedicine, Washington DC, USA.
- 2015 Efficient Algorithms for Finding Edit Distance Based Motifs. NPAW Postdoc Research Datablitz Presentation Competition, University of Connecticut, Storrs, USA.
- 2013 Scheduling Light-trails on WDM Rings. Indian Institute of Technology – Roorkee, India.
- 2011 Recovery of a Sparse Integer Solution to an Underdetermined System of Linear Equations. NeurIPS Sparse Representation and Low-rank Approximation Workshop, Granada, Spain.
- 2011 [Poster] Phase Identification in Smart Grids. IBM India Research Lab, Bangalore, India.
- 2009 Scheduling Light-trails on WDM Rings. International Conference on Advanced Computing and Communications (ADCOM), Bangalore, India.
- 2004 Design and Validation Techniques to Implement a Robust Hindsight Feature on ISA Simulators. Global Signal Processing Expo and Conference, Santa Clara, USA.
- 2004 A methodology for Saving and Restoring the State of ISA simulators. Global Signal Processing Expo and Conference, Santa Clara, USA.
- 2003 TMS5xxx DSP Architecture and ISA Simulator in TI Code Composer Studio. Osmania University, Hyderabad, India.
- 2000 On Augmenting Senses by Digitizing Smell. IDEON Idea Presentation Competition, Indian Institute of Technology – Kharagpur, India.

Professional Services and Membership

Journal Reviewer

- 2025 Nature Communications
- 2024 Nature Communications; Cell Reports; Investigative Ophthalmology & Visual Science; BMC Medical Genomics.
- 2023 Nature Communications; Nature Genetics; Investigative Ophthalmology & Visual Science (IOVS); Proceedings of the National Academy of Sciences (PNAS).
- 2022 (Oxford) Bioinformatics; Molecular Ecology.
- 2021- Review Editor for Computational Genomics, Frontiers in Genetics.
- 2021 Frontiers in Genetics; (Oxford) Bioinformatics.
- 2020 Nature Scientific Report; Computer Methods in Biomechanics and Biomedical Engineering (CMBBE); BMC Biology.
- 2019-20 BMC Supplements.
- 2019 IEEE/ACM Transactions on Computational Biology and Bioinformatics (TCBB); PLoS Computational Biology.
- 2015 Journal of Grid Computing.
- 2011 Theoretical Computer Science (TCS).

Conference External Reviewer

- 2025,26 ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB).
2019 Intelligent Systems for Molecular Biology / European Conference on Computational Biology (ISMB/ECCB).
2018-21,25 International Conference on Research in Computational Molecular Biology (RECOMB).
2016 Indian Control Conference (ICC).
2014-15 IEEE Intl. Conference on Computational Advances in Bio and medical Sciences (ICCABS).

Conference / Symposium / Workshop Organizing Committee

- 2026 Program Committee Member, First Staff Scientist Symposium, NIH Research Festival.
2020- Program Committee Member, Workshop on Graphs and more Complex structures for Learning and Reasoning (GCLR), AAAI Conference on Artificial Intelligence.
2015-16 IEEE Intl. Conference on Computational Advances in Bio and medical Sciences (ICCABS).
2014 Parallel Computing Architecture and Applications on Multi-Core to Many-Core Processing Systems (PCAMS 2014), Indian Institutes of Technology – Bombay and Centre for Development of Advanced Computing (C-DAC), Pune, India.
2011 Foundations of Software Technology and Theoretical Computer Science (FSTTCS), Indian Institute of Technology – Bombay, Mumbai, India.

Other Service / Membership

- 2023- Member of Association for Research in Vision and Ophthalmology (ARVO).
2020- Member of the Genetics Society of America (GSA).
2018- Member of the International Society for Computational Biology (ISCB).